

# Mock Midterm 1

EECS 245, Winter 2026 at the University of Michigan

Name: \_\_\_\_\_

username: \_\_\_\_\_

UMID: \_\_\_\_\_

Room: ☐ 1014 DOW    ☐ At Home

## Instructions

- This exam consists of 7 questions. **On the real midterm, we will also state the total number of points for each question. We make no guarantees on the number of questions, points, or specific questions on the real midterm.**
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page in the space provided.
- For free response problems, you must show all of your work (unless otherwise specified), and circle your final answer. We will not grade work that appears elsewhere, and you may lose points if your work is not shown.
- For multiple choice problems, completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.
  - ☐ A bubble means that you should only select one choice.
  - ☐ A square box means you should select all that apply.
- You may refer to a single two-sided handwritten notes sheet. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

You are to abide by the University of Michigan/Engineering Honor Code. To receive a grade, please sign below to signify that you have kept the Honor Code pledge.

*I have neither given nor received aid on this exam, nor have I concealed any violations of the Honor Code.*

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### Problem 1: Doubling Down

Suppose we'd like to find the optimal parameter,  $w^*$ , for the constant model  $h(x_i) = w$ , given a dataset of  $n$  points  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ . To do so, we use the **doubly squared** loss function,  $L_{\text{ds}}$ , defined below.

$$L_{\text{ds}}(y_i, w) = (y_i^2 - w^2)^2$$

- a) Find  $\frac{dR_{\text{ds}}}{dw}$ , the derivative of average doubly squared loss (i.e. the empirical risk) with respect to  $w$ .

- b) Show that the value of  $w$  that minimizes average doubly squared loss is

$$w^* = \sqrt{\frac{1}{n} \sum_{i=1}^n y_i^2}$$

## Problem 2: Absolutely...

Consider a dataset of 3 values,  $y_1 < y_2 < y_3$ , with a mean of 2. Let

$$Y_{\text{abs}}(w) = \frac{1}{3} \sum_{i=1}^3 |y_i - w|$$

represent the mean absolute error of a constant prediction  $w$  on this dataset of 3 values.

Similarly, consider another dataset of 5 values,  $z_1 < z_2 < z_3 < z_4 < z_5$ , with a mean of 12. Let

$$Z_{\text{abs}}(w) = \frac{1}{5} \sum_{i=1}^5 |z_i - w|$$

represent the mean absolute error of a constant prediction  $w$  on this dataset of 5 values.

Suppose that  $y_3 < z_1$ , and that  $T_{\text{abs}}(w)$  represents the mean absolute error of a constant prediction  $w$  on the combined dataset of 8 values,  $y_1, y_2, y_3, z_1, z_2, z_3, z_4, z_5$ .

a) Fill in the blanks to complete the sentence:

---(i)--- minimizes  $Y_{\text{abs}}(w)$ , ---(ii)--- minimizes  $Z_{\text{abs}}(w)$ , and ---(iii)--- minimizes  $T_{\text{abs}}(w)$ .

Note that in the options below,  $[a, b]$  represents the range of values between  $a$  and  $b$ , including both  $a$  and  $b$ .

- (i) ☐  $y_1$    ☐ any value in  $[y_1, y_2]$    ☐  $y_2$    ☐  $y_3$    ☐  $z_1$
- (ii) ☐  $z_1$    ☐  $z_2$    ☐ any value in  $[z_2, z_3]$    ☐ any value in  $[z_3, z_4]$    ☐  $z_3$
- (iii) ☐  $y_2$    ☐  $y_3$    ☐ any value in  $[y_3, z_1]$    ☐ any value in  $[z_1, z_2]$   
☐ any value in  $[z_2, z_3]$

b) For any value  $w$ , it's true that

$$T_{\text{abs}}(w) = \alpha Y_{\text{abs}}(w) + \beta Z_{\text{abs}}(w)$$

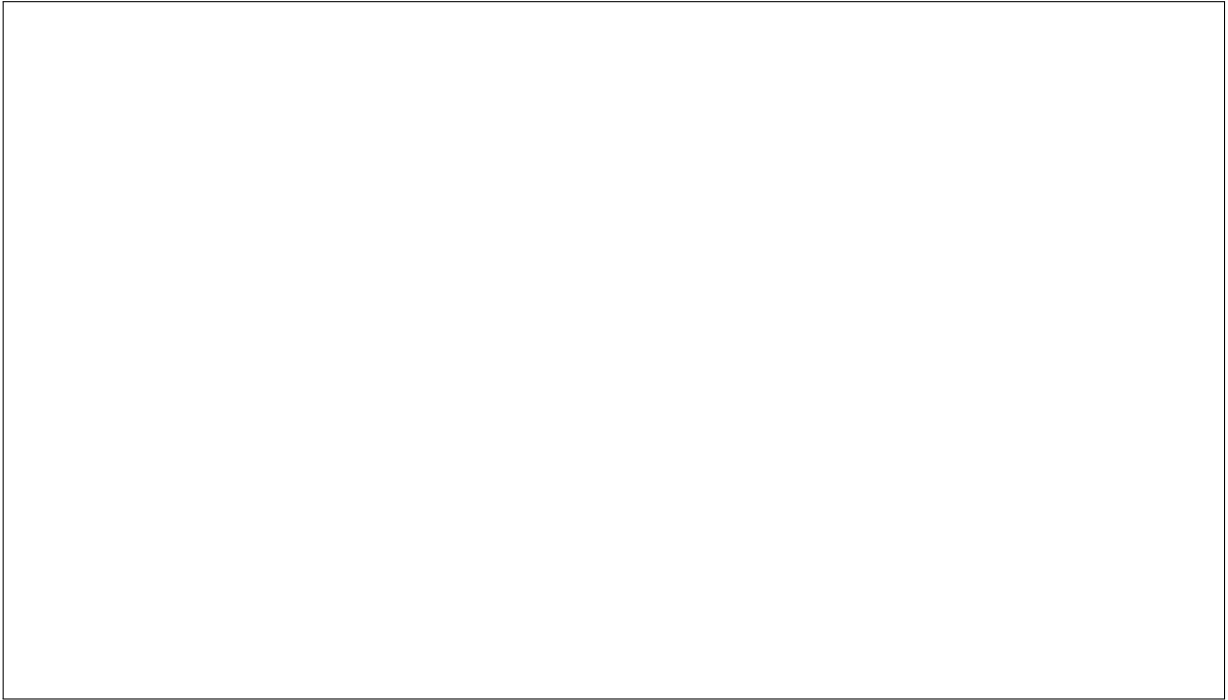
for some constants  $\alpha$  and  $\beta$ . Determine the values of  $\alpha$  and  $\beta$ . Both answers should be integers or simplified fractions with no variables.

$$\alpha = \boxed{\phantom{000000}} \qquad \beta = \boxed{\phantom{000000}}$$

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- c) Show that  $Y_{\text{abs}}(z_1) = z_1 - 2$ .

*Hint: Use the fact that you know the mean of  $y_1, y_2, y_3$ .*



- d) Suppose the minimum possible **output** of  $T_{\text{abs}}(w)$  in the full dataset of 8 values is 6. What is the value of  $z_1$ ?

*Hint: You'll need to use the answers to the previous parts.*

☐ -2   ☐ 0   ☐ 2   ☐ 3   ☐ 5   ☐ 6   ☐ 7   ☐ 9

### Problem 3: Switcheroo

Consider the following datasets, both consisting of  $n = 8$  points.

- "Old" dataset:  $(3, 8), (7, 2), (x_3, y_3), \dots, (x_8, y_8)$
- "New" dataset:  $(3, 2), (7, 8), (x_3, y_3), \dots, (x_8, y_8)$

Note that the only difference between the datasets is that the first two  $y$ -values have been swapped.

a) Which of the following quantities **are guaranteed to be different** between the old and new datasets? Select all that apply.

- ☐ The mean of the  $x$ -values,  $\bar{x}$
- ☐ The mean of the  $y$ -values,  $\bar{y}$
- ☐ The variance of the  $x$ -values,  $\sigma_x^2$
- ☐ The variance of the  $y$ -values,  $\sigma_y^2$
- ☐ The correlation coefficient between the  $x$ -values and the  $y$ -values,  $r$

b) Let  $m_{\text{old}}$  and  $m_{\text{new}}$  be the slopes of the regression lines fit to the old and new datasets, respectively. Given that  $\sigma_x^2 = 50$ , find the value of  $m_{\text{new}} - m_{\text{old}}$ . Your final answer should be a number with no variables.

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**Problem 4: Variability**

Let  $\vec{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$  and  $y = \begin{bmatrix} 3 \\ c \end{bmatrix}$ , where  $c \in \mathbb{R}$  is a constant.

a) Fill in the blanks to complete the sentence:

The value of  $c$  that makes  $|\vec{x} \cdot \vec{y}|$  as small as possible is \_\_\_(i)\_\_\_; when using that value of  $c$ ,  $\vec{x}$  and  $\vec{y}$  are \_\_\_(ii)\_\_\_.

(i)

(ii)

(1-3 words)

b) Suppose the projection of  $\vec{y}$  onto  $\vec{x}$  is  $\begin{bmatrix} -12/5 \\ 9/5 \end{bmatrix}$ , for some value of  $c$ . What is the value of  $c$ ? Show your work, and circle your final answer, which should be a number with no variables.

As a refresher,  $\vec{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$  and  $\vec{y} = \begin{bmatrix} 3 \\ c \end{bmatrix}$ , where  $c \in \mathbb{R}$  is a constant.

In the next two parts, suppose  $\theta_c$  is the angle between  $\vec{x}$  and  $\vec{y}$ . As  $c$  gets larger and larger,  $\cos \theta_c$  gets closer and closer to  $L$ , i.e.

$$\lim_{c \rightarrow \infty} \cos \theta_c = L$$

c) What is the value of  $L$ ?

- ☐  $3/4$    ☐  $-3/4$    ☐  $3/5$    ☐  $-3/5$    ☐  $4/5$    ☐  $-4/5$    ☐ None of these

d)  $\cos^{-1}(L)$  is also equal to the angle between  $\vec{x}$  and a particular unit vector,  $\vec{u}$ . Find  $\vec{u}$  and explain your answer.

*Hint: This is more of a conceptual question than a computational one.*



**Problem 5: Well, It Depends**

Let  $\vec{u} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 4 \end{bmatrix}$ ,  $\vec{v} = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 6 \end{bmatrix}$ , and  $\vec{w} = \begin{bmatrix} 1 \\ -1 \\ \alpha \\ 2 \end{bmatrix}$ , where  $\alpha \in \mathbb{R}$  is a constant.

In parts **a)**, **b)**, **c)**, and **d)**, suppose  $\alpha = 3$ . Fill in the blanks to complete each sentence.

**a)** The two vectors  $\{\vec{w}, \vec{u} - \vec{w}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

**b)** The two vectors  $\{\vec{v}, 2\vec{v}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

**c)** The three vectors  $\{\vec{u}, \vec{v}, \vec{w}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

**d)** The four vectors  $\{\vec{u}, \vec{v}, \vec{w}, \vec{w} - \vec{u}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

Now, suppose  $\alpha = -2$ . Fill in the blanks to complete each sentence.

**e)** The three vectors  $\{\vec{u}, \vec{v}, \vec{w}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

**f)** The four vectors  $\{\vec{u}, \vec{v}, \vec{w}, \vec{w} - \vec{u}\}$  are \_\_\_(i)\_\_\_, and span a \_\_\_(ii)\_\_\_-dimensional subspace of  $\mathbb{R}^4$ .

(i) ☐ linearly independent ☐ linearly dependent

(ii) ☐ 1 ☐ 2 ☐ 3 ☐ 4

### Problem 6: Covering All the Bases

- a) Find a basis for the following subspace of  $\mathbb{R}^3$ :

$$S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid 6x - y + z = 0 \right\}$$

- b) The equations  $6x - y + z = 0$  and  $4x + 4y = 0$  intersect in a line in  $\mathbb{R}^3$ . Find the equation of this line in **parametric** form.

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### Problem 7: Catchy

Suppose  $\vec{x} \in \mathbb{R}^n$ . Prove that the  $L_1$  norm of a vector is less than or equal to  $\sqrt{n}$  times the  $L_2$  norm of a vector, i.e.

$$\|\vec{x}\|_1 \leq \sqrt{n}\|\vec{x}\|_2$$

*Hint: Use the Cauchy-Schwarz inequality, which states that for any two vectors  $\vec{u}, \vec{v} \in \mathbb{R}^n$ ,  $|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\|_2 \|\vec{v}\|_2$ . Most of your job is to choose the right vectors  $\vec{u}$  and  $\vec{v}$  to apply the Cauchy-Schwarz inequality to.*